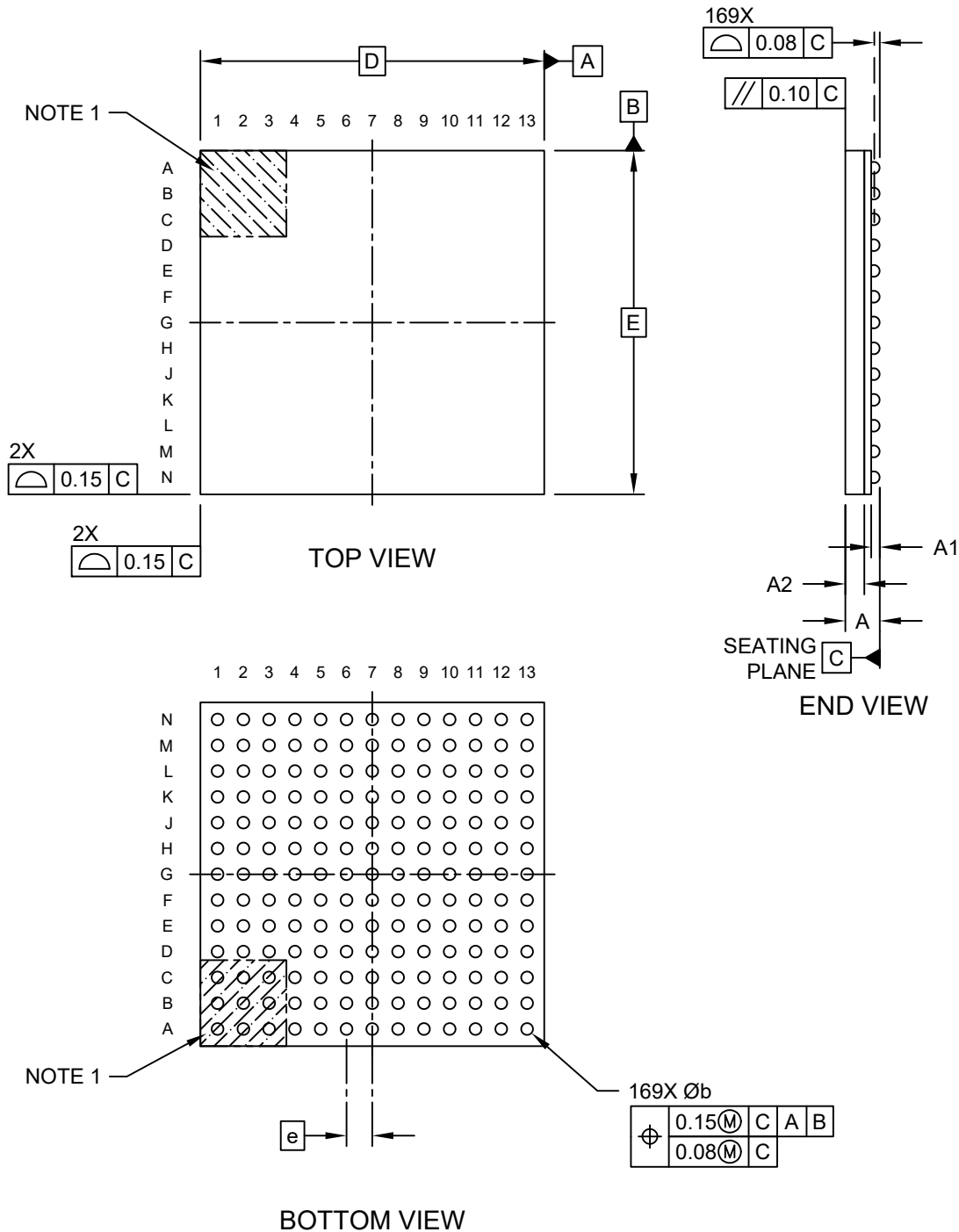


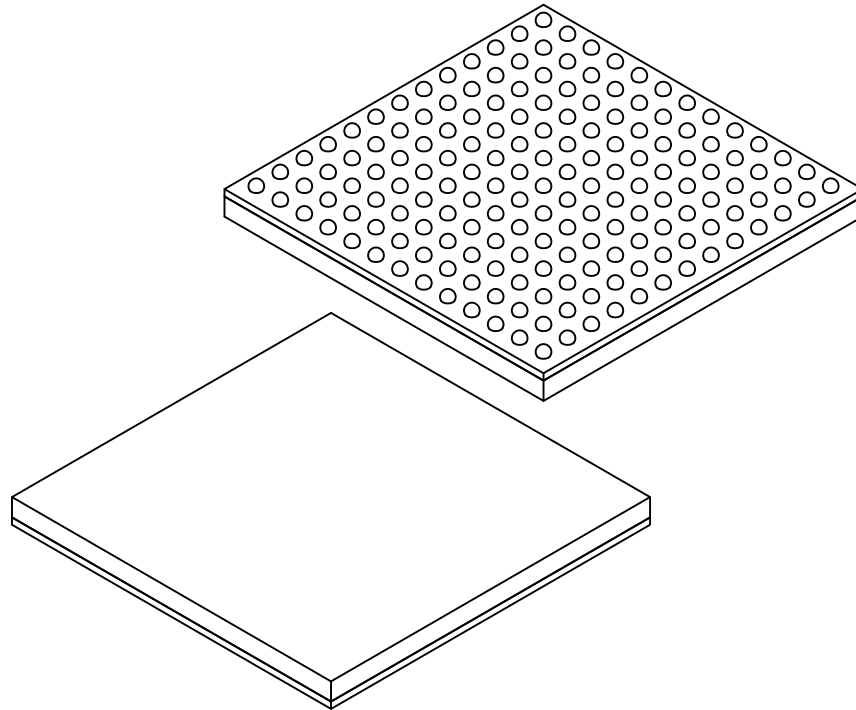
169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
(Complies with JEDEC Terminal Assignment recommendations)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
(Complies with JEDEC Terminal Assignment recommendations)**

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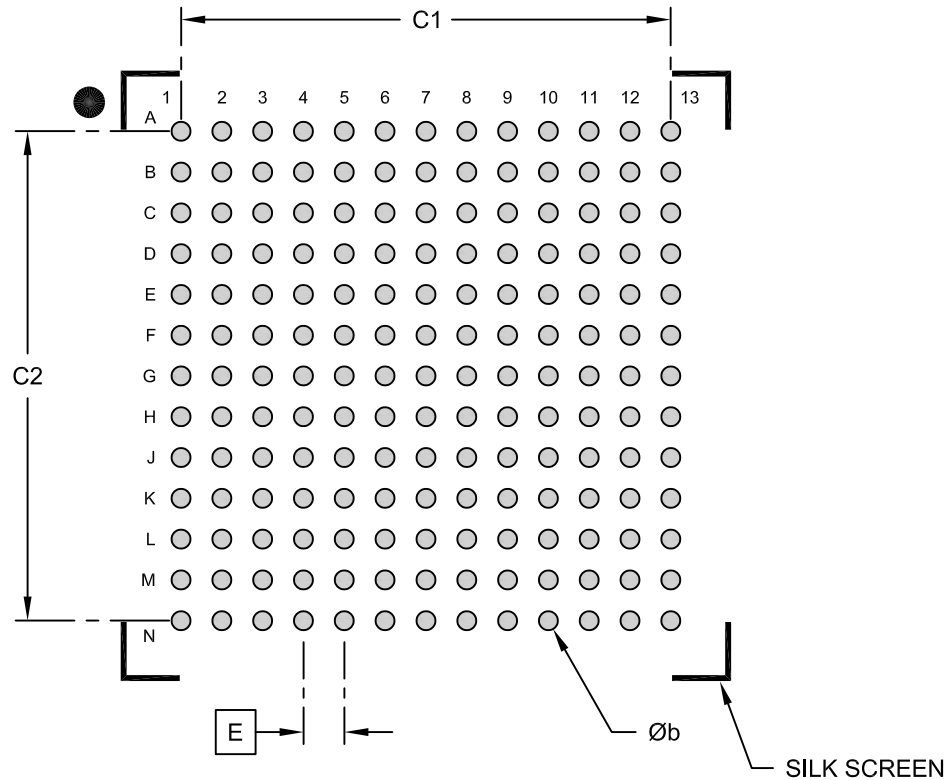
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	169		
Pitch	e	0.75 BSC		
Overall Height	A	–	–	1.10
Ball Height	A1	0.21	0.32	–
Mold Thickness	A2	0.40	0.45	0.50
Overall Length	D	10.00 BSC		
Overall Width	E	10.00 BSC		
Ball Diameter	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

**169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
(Complies with JEDEC Terminal Assignment recommendations)**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.75 BSC		
Contact Pad Spacing	C1		9.00	
Contact Pad Spacing	C2		9.00	
Contact Pad Diameter (X169)	b		0.35	

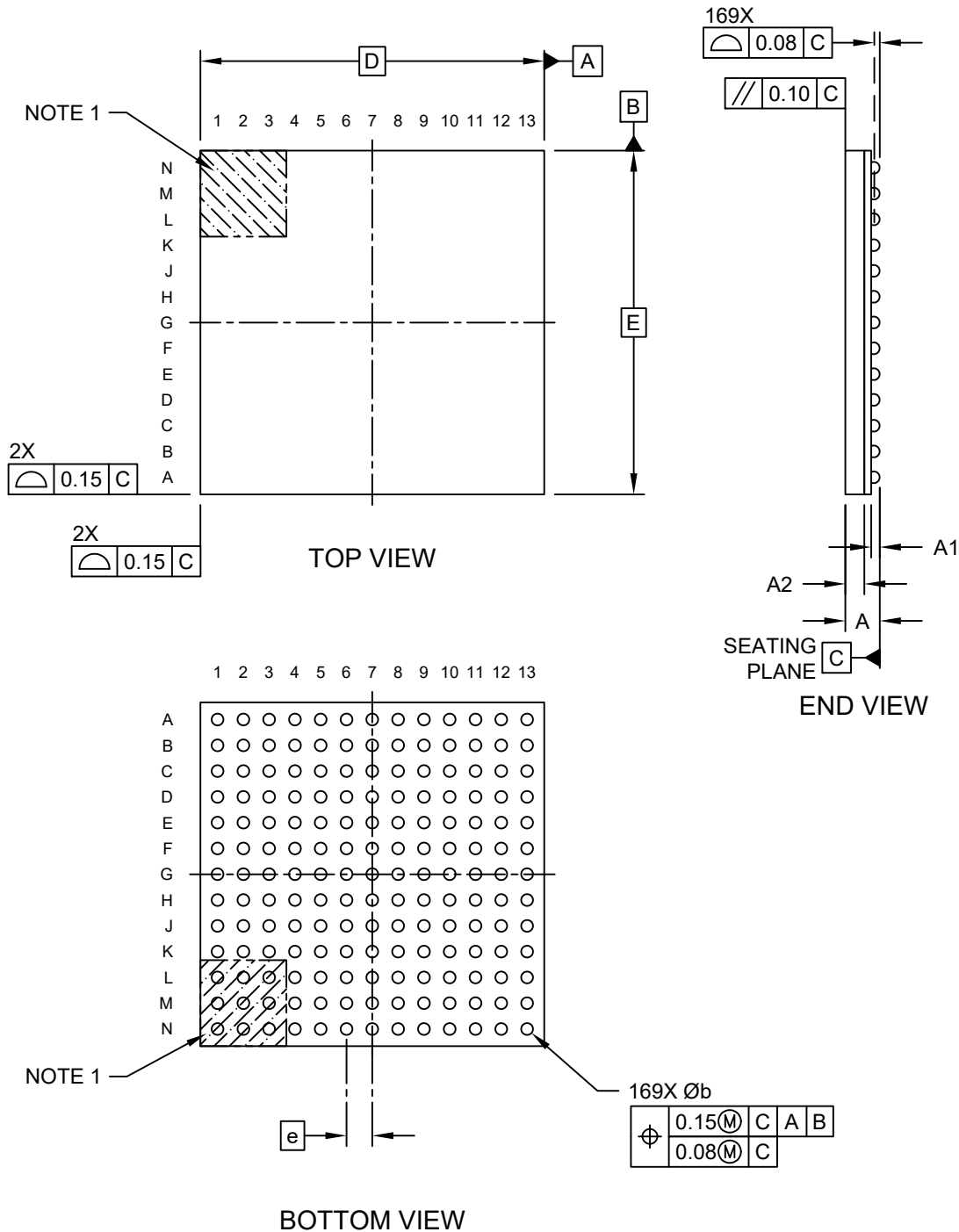
Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

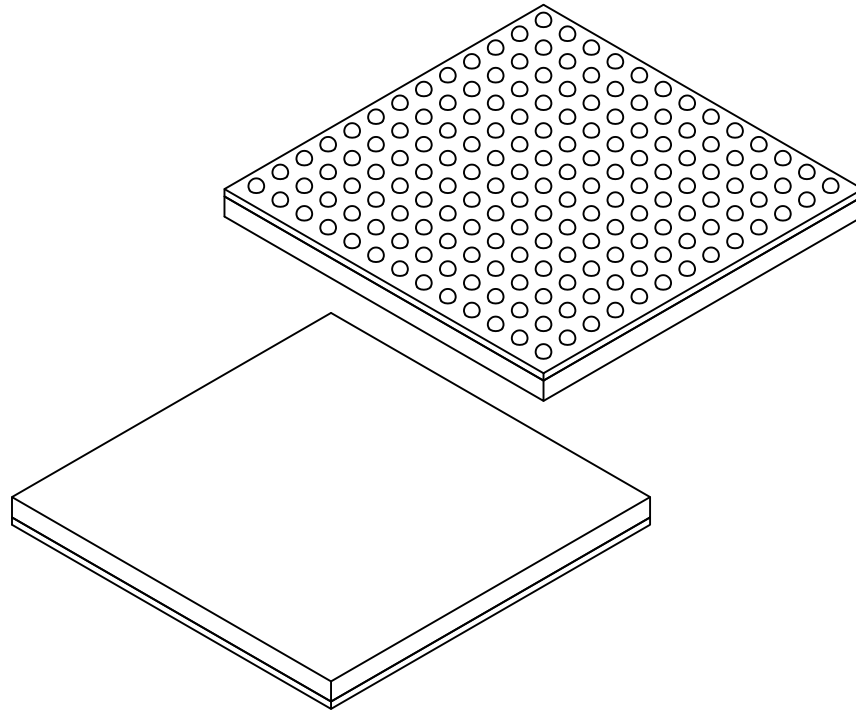
169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
Alternate Terminal Assignments (does not comply with JEDEC recommendations)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
Alternate Terminal Assignments (does not comply with JEDEC recommendations)

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



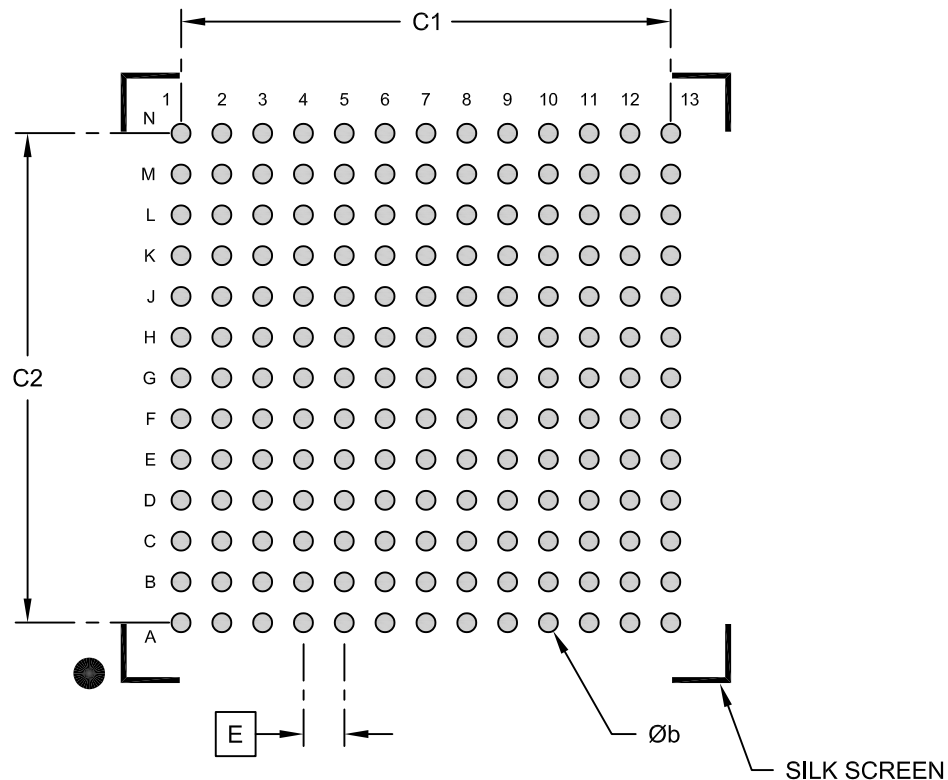
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	169		
Pitch	e	0.75 BSC		
Overall Height	A	–	–	1.10
Ball Height	A1	0.21	0.32	–
Mold Thickness	A2	0.40	0.45	0.50
Overall Length	D	10.00 BSC		
Overall Width	E	10.00 BSC		
Ball Diameter	b	0.35	0.40	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
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169-Ball Thin Fine Pitch Ball Grid Array (7G) - 10x10x1.10 mm Body [TFBGA]
Alternate Terminal Assignments (does not comply with JEDEC recommendations)

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.75 BSC		
Contact Pad Spacing	C1		9.00	
Contact Pad Spacing	C2		9.00	
Contact Pad Diameter (X169)	b		0.35	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

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